

The ASEAN EV Triangle: Thailand, Indonesia, Vietnam in a Three-Way Race for \$200B in EV FDI

Blue Lotus Research Intelligence | August 2026

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I. Opening: Three Nations, One Crown

In the summer of 2026, three Southeast Asian nations are engaged in the most consequential industrial competition in the region's history. Thailand, Indonesia, and Vietnam are each racing to capture the defining economic prize of the decade: becoming the electric vehicle manufacturing hub of ASEAN, a market of 680 million people and the world's fifth-largest automotive sub-region. The prize is not abstract. It is measured in factories, jobs, technology transfer, and a seat at the table of global automotive supply chains that are being rewired at a speed not seen since the original Japanese transplant wave of the 1970s and 1980s.

The numbers frame the urgency. BEV penetration in ASEAN-6 countries reached 21 percent of total car sales in the first half of 2026, nearly double the 12 percent rate recorded just a year earlier. [Source: TNGlobal / BMI, July 2026 — <https://technode.global/2026/07/17/asean-emerges-as-key-market-manufacturing-hub-for-evs-bmi/>] Thailand has secured over \$4.1 billion in EV supply chain investment pledges as of mid-2026. [Source: Malay Mail / TNGlobal, July 2026 — <https://technode.global/2026/07/06/thailand-secures-4-1b-in-ev-supply-chain-investment-featuring-investments-from-china-korea-japan/>] Indonesia is hosting a \$6 billion CATL battery integration megaproject. [Source: ChinaEVHome, June 2026 — <https://chinaevhome.com/2025/06/30/catl-breaks-ground-on-6-billion-battery-plant-in-indonesia-production-scheduled-for-late-2026/>] Vietnam's national champion VinFast delivered 175,099 vehicles domestically in 2025 and is targeting 300,000 global deliveries in 2026. [Source: TNGlobal, January 2026 — <https://technode.global/2026/01/16/vinfast-delivers-175099-evs-in-vietnam-in-2025/>]

Yet for all three nations, the structural advantage of winning is enormous — and only partially about cars. The country that locks in the deepest EV supply chain integration will dictate downstream industrial policy, labour markets, and technology capability for a generation. The country that fails to attract anchor investments risks deindustrialisation as Chinese OEMs bypass it entirely, routing production through whichever jurisdiction offers the best combination of incentives, logistics, and

local content flexibility.

The race is real. The stakes are enormous. The outcome is not yet determined. This report dissects the competitive positioning of each nation, maps the strategic logic of the Chinese OEMs driving the investments, assesses Japan's rearguard defence, and renders a superforecast on the winner by 2030.

II. The Prize: ASEAN's EV Market Opportunity

The Automotive Foundation

ASEAN's automotive market is substantial but structurally polarised. The region sold approximately 3.25 million vehicles in 2025, roughly flat from 3.26 million in 2024. [Source: Focus2move — <https://www.focus2move.com/asean-vehicle-market/>] Five economies — Indonesia, Thailand, Malaysia, the Philippines, and Vietnam — account for the vast bulk of sales. Indonesia is the largest single market by volume, Thailand the largest by production, and Vietnam the fastest-growing in EV penetration terms.

EV Penetration: A Structural Inflection

The shift to EVs is not gradual — it is compressing. BEV sales across ASEAN-6 have gone from marginal in 2022 to a 21 percent share in H1 2026. The country-level divergence is stark. Vietnam leads in penetration at approximately 35 percent, driven entirely by VinFast's domestic dominance. Singapore, constrained by vehicle quota policies, reaches 62 percent but represents tiny absolute volumes. Thailand and Indonesia range between 22 and 30 percent BEV market share in H1 2026. Malaysia trails at around 6 percent. [Source: TNGlobal / Ember, August 2026 — <https://technode.global/2026/08/27/southeast-asias-ev-sales-accelerate-as-energy-crisis-continues/>]

Vietnam and Thailand are forecast to jointly account for more than 72 percent of ASEAN passenger EV sales in 2026. Thailand's share of total ASEAN EV volume stands at approximately 38 percent, Vietnam at 29.5 percent, and Indonesia at 19.5 percent, with the remainder distributed across ASEAN. [Source: Mordor Intelligence — <https://www.mordorintelligence.com/industry-reports/asean-electric-vehicle-market>]

Market Size and 2030 Trajectory

The ASEAN EV market is valued at approximately \$5.99 billion in 2026 according to one set of projections, growing at a 31.55 percent compound annual growth rate to reach \$23.58 billion by 2031. [Source: GMInsights — <https://www.gminsights.com/industry-analysis/asean-electric-vehicle-market>] Alternative projections are more bullish, placing the 2026 ASEAN EV market at \$18.7 billion. ASEAN-6 governments have collectively committed over \$3.5 billion in direct subsidies, tax exemptions, and infrastructure co-investments to catalyse EV adoption through the end of the decade.

The 2030 horizon matters enormously because that is when EV penetration in key markets is projected to reach a tipping point. Thailand's official target is that 30 percent of vehicles manufactured domestically by 2030 will be zero-emission — the "30@30" strategy — targeting 1.123 million ZEVs annually. Indonesia is targeting its own domestic EV ramp-up with nickel-integrated battery production fully operational before 2028. Vietnam has no formal ZEV production target but VinFast's trajectory implies 300,000-plus vehicles annually by 2027-2028.

Chinese Brands vs Japanese Incumbents

The competitive narrative in ASEAN is fundamentally about Chinese displacement of Japanese incumbents, and the speed of that displacement is accelerating faster than most incumbents forecast. In Thailand, 85 percent of EV sales in 2024 were Chinese-made. [Source: Global Voices — <https://globalvoices.org/2025/10/09/how-chinese-automakers-are-reshaping-the-ev-landscape-in-southeast-asia/>] Chinese EV brands' sales in Southeast Asia's top four markets rose 58 percent in Q1 2026 year-on-year. [Source: Just-Auto — <https://www.just-auto.com/features/chinese-sales-in-se-asias-top-4-markets-rise-58-in-q1/>]

The strategic logic from the Chinese side is explicit: ASEAN markets that do not yet have the tariff walls facing Chinese EVs in the US and EU are the primary export springboard for China's next wave of global automotive expansion. Chinese companies plan to produce more than one million vehicles in ASEAN countries in 2026, with approximately 600,000 expected to be EVs.

ASEAN as Export Platform

Australia and Saudi Arabia are the two largest automotive export destinations for ASEAN outside the region. [Source: IEA Global EV Outlook 2026 — <https://www.iea.org/reports/global-ev-outlook-2026/manufacturing-and-trade>] As Chinese-built EVs face tariffs in the US, EU, Canada, Turkey, and Brazil, ASEAN-produced vehicles — manufactured by Chinese OEMs under ASEAN certificates of origin — present a strategic circumvention pathway that is simultaneously an opportunity for host-country job creation and a strategic risk for Western policymakers. The Middle East and Africa EV market is valued at \$5.06 billion in 2026 growing at 32.15 percent CAGR, representing a natural downstream export destination for ASEAN-produced EVs over the 2027-2030 horizon.

III. Thailand: The Established Platform

The Incumbent Advantage

Thailand enters the EV race with the region's deepest automotive industrial base. The country's existing manufacturing ecosystem — 1,800 automotive parts suppliers, established logistics corridors, and five decades of Tier 1 OEM relationships — gives it structural advantages that neither Indonesia nor Vietnam can replicate on a five-year timeline. Thailand currently exports more than half of its total vehicle production, a logistics and trade infrastructure achievement that represents

enormous embedded capital. [Source: EV Magazine — <https://evmagazine.com/news/thailand-a-global-hub-for-electric-vehicle-production>]

Thailand's automotive sector produces approximately 1.7 to 1.8 million vehicles annually and is the twelfth-largest automotive producer globally. The transition of even a portion of this base to EV production is therefore of a different order of magnitude than building greenfield capacity in Indonesia or Vietnam.

The EV3.5 Policy Engine

Thailand's Board of Investment EV3.5 policy is the region's most sophisticated EV incentive architecture. From 2026, investors must assemble two vehicles in Thailand for every one imported under subsidy conditions; by 2027, the ratio rises to three-to-one. [Source: ASEAN Briefing — <https://www.aseanbriefing.com/news/how-boi-ev-3-5-shapes-the-future-of-ev-battery-investments-in-thailand/>] Critically, since mid-2025, the government has allowed exports of locally built EVs to count toward these localisation obligations — a strategic concession that gives Chinese OEMs using Thailand as a regional export hub the flexibility to balance domestic sales with ASEAN and Middle East shipments.

Thailand secured over \$4.1 billion in EV supply chain investment across 2025 and the first half of 2026, with investments from China, Korea, and Japan. [Source: TNGlobal, July 2026 — <https://technode.global/2026/07/06/thailand-secures-4-1b-in-ev-supply-chain-investment-featuring-investments-from-china-korea-japan/>] In December 2025, the BOI added hybrid and mild-hybrid vehicles to the incentive scope, a tactical move designed to retain Japanese OEM investment. This is a double-edged concession: it broadens the investment tent but dilutes the pure-EV industrial strategy.

The Chinese Invasion of Rayong and the EEC

The Eastern Economic Corridor (EEC) has become the locus of Chinese automotive capital in ASEAN. At least five Chinese automotive brands secured BOI approval for assembly operations between 2022 and 2025, with collective announced capacity exceeding 600,000 units per year. [Source: Sinocities Substack — <https://sinocities.substack.com/p/evs-batteries-and-data-centers-the>]

BYD opened its first ASEAN assembly plant in Thailand in July 2024 at the WHA Eastern Seaboard Industrial Estate in Rayong, with designed capacity of 150,000 vehicles per year. [Source: TopGear Philippines — <https://www.topgear.com.ph/news/industry-news/byd-thailand-plant-opening-a5100-20240705>]

Great Wall Motor acquired the mothballed General Motors assembly plant in Rayong for its ORA and Haval electric and hybrid lines, with capacity around 80,000 units per year. SAIC Motor and Changan Automobile have both established domestic assembly lines. Changan launched manufacturing operations in 2025 with a 100,000-unit facility. As of mid-2026, Hyundai Mobility launched production in Thailand, and Omoda and Jaecoo commenced 2026 production rollouts.

Battery Assembly Without Cell Production

Thailand's critical Achilles' heel in the EV race is battery cell production. Despite massive assembly investment, Thailand has not attracted significant lithium-ion cell manufacturing. Assembly plants using imported Chinese battery packs represent value-added at the lower end of the supply chain spectrum. The BOI's EV battery investment push aims to attract cathode material producers and cell manufacturers, but as of mid-2026, Thailand remains an assembly-dominant economy in battery terms. This is strategically significant: the highest-value layers of the EV supply chain — cell chemistry, battery management systems, and module integration — remain predominantly in China.

Japan's Hybrid Defence

Toyota, Honda, and Mitsubishi continue to dominate Thailand's conventional vehicle market, and the December 2025 BOI extension of incentives to hybrids represents a Japanese lobby victory. Japan's "multi-pathway" strategy is well-suited to Thailand's incremental transition because it allows Japanese OEMs to defend market share with hybrid products while delaying full BEV investment. However, this defence is time-limited: as Chinese BEVs extend their price-to-quality advantage and Thai consumer preferences shift, the hybrid buffer will erode.

Thailand's Export Target and Strategic Position

Thailand's 30@30 strategy — 30 percent ZEV manufacturing by 2030, targeting 1.123 million zero-emission vehicles annually — positions the country as a production-for-export model. More than half of Thailand's vehicle production is exported, and the EV transition preserves this export orientation. The target markets — Australia, ASEAN neighbours, Middle East — are reachable. The risk is that Chinese OEMs, having received BOI incentives, optimise for their own global supply chains rather than Thailand's national industrial interests.

IV. Indonesia: The Resource Play

The Nickel Thesis

Indonesia's EV strategy rests on a single geopolitical asset of enormous potential value: the world's largest nickel reserves. Indonesia is on track to account for approximately 60 percent of global nickel supply by 2025, rising to an estimated 75 percent by 2030. [Source: Fastmarkets — <https://www.fastmarkets.com/insights/feoc-definition-leaves-indonesian-nickel-outside-ira-tax-credits-andrea-hotter/>] The strategic vision is vertical integration from ore in the ground to battery cell to EV assembly — capturing the full value chain on Indonesian soil rather than exporting raw ore to Chinese processors who capture the margin.

This thesis is intellectually compelling. Nickel is the critical input to nickel-manganese-cobalt (NMC) battery chemistry, and Indonesia's resource endowment gives it a structural cost advantage in NMC battery production that no other ASEAN nation can match. If Indonesia can execute the vertical

integration from ore to cell, it would occupy the most defensible position in the ASEAN EV supply chain.

CATL's \$6 Billion Bet

The centrepiece of Indonesia's battery strategy is the CATL Indonesia Battery Integration Project, a \$6 billion commitment to build a complete value chain facility in Karawang, West Java. The project covers nickel mining and processing, battery materials production, battery cell manufacturing, and recycling. [Source: CATL official announcement — <https://www.catl.com/en/news/6481.html>] The initial stage targets 6.9 GWh annual production capacity by end-2026 — enough to power approximately 100,000 EVs. Phase two expands capacity to over 15 GWh by the end of the decade.

Indonesian President Prabowo Subianto attended the CATL groundbreaking ceremony in Karawang in June 2026, underscoring the political commitment at the highest level. The project is expected to create 8,000 direct jobs and 35,000 indirect employment opportunities. [Source: Battery Tech Online — <https://www.batterytechonline.com/battery-manufacturing/catl-launches-6b-indonesia-battery-integration-project-for-ev-ecosystem-growth/>]

The Hyundai-LG Energy Solution joint venture adds another pillar: in July 2024, the partnership inaugurated Southeast Asia's first EV battery cell plant in Karawang with 10 GWh annual output — a \$1.1 billion investment supplying cells for the Ioniq 5, Kona EV, and Kia models. [Source: Indonesia EV Ecosystem Bulletin — <https://ibc-bulletin-vol4.vercel.app/>]

BYD's Indonesia Manufacturing Hub

BYD's Indonesia factory, representing approximately \$1.3 billion in investment, commenced production in January 2026. The West Java facility has annual capacity of approximately 150,000 vehicles — making hybrids and EVs including the Atto 3, Dolphin, and Seal — and is positioned as a right-hand-drive export hub for ASEAN, the Middle East, and Africa. BYD targets a localisation rate of 50 to 60 percent, which aligns with Indonesia's domestic content requirements. [Source: Electric Vehicles News — <https://electric-vehicles.com/byd/byds-1-billion-indonesia-plant-on-track-for-q3-production-start/>]

Prabowo's Policy Architecture

President Prabowo's EV policy combines demand stimulation with supply-chain nationalism. VAT on EV purchases has been cut from 11 percent to 1 percent for locally manufactured EVs with at least 40 percent local content. Luxury sales tax exemptions and import duty waivers — contingent on OEMs committing to local production — have driven the assembly investment wave. [Source: GMInsights — <https://www.gminsights.com/industry-analysis/asean-electric-vehicle-market>]

In April 2025, Prabowo's administration raised nickel ore royalties from a flat 10 percent to a sliding scale of 14 to 19 percent depending on nickel benchmark prices. This "resource to industry" policy captures more resource rent for the state while maintaining the incentive structure for downstream processing and battery manufacturing.

The Domestic Market Imperative

Indonesia's 270 million population is its most powerful structural argument in the EV race. The domestic market alone — if EV penetration reaches 30 percent of total sales by 2030 — represents approximately 750,000 to 900,000 EV units annually at current market run-rates. No other ASEAN EV market offers this scale of domestic absorption. Indonesian OEMs and government can use domestic market access as leverage in negotiations with Chinese, Korean, and Japanese investors in ways that Thailand and Vietnam cannot.

Indonesia's Q2 2026 performance is instructive: EV boom dynamics are visible, with the country's automotive sector surging 34 percent in that period. [Source: KR-Asia — <https://kr-asia.com/ev-boom-drives-asean-car-sales-in-q2-with-indonesia-surging-34>] BYD captured approximately 5.3 percent of Indonesia's overall vehicle market in H1 2026.

The FEOC Complication

Indonesia's nickel vertical integration thesis faces a structural complication that threatens US market access. The Inflation Reduction Act's Foreign Entity of Concern (FEOC) provisions bar Chinese-majority joint ventures from IRA clean vehicle tax credit eligibility. The bulk of Indonesian nickel processing involves Chinese joint-venture partners — normally Indonesian in the mining phase, Chinese in the smelting and refining phases. [Source: Fastmarkets FEOC analysis — <https://www.fastmarkets.com/insights/feoc-definition-leaves-indonesian-nickel-outside-ira-tax-credits-andrea-hotter/>]

This means that batteries manufactured in Indonesia using nickel processed by Chinese-majority JVs are ineligible for IRA section 30D credits, effectively closing the US market to Indonesian-manufactured EVs destined for American consumers. As of 2026, a single FEOC-linked supplier three tiers down the supply chain can disqualify an entire battery project from federal clean energy credits. [Source: ArentFox Schiff / Certainty Software — <https://www.certaintysoftware.com/feoc-foreign-entity-of-concern/>] Indonesia is actively negotiating a bilateral critical minerals agreement with the US to address this, but resolution remains outstanding as of the time of this report.

V. Vietnam: The Latecomer with Ambition

VinFast as National Champion

Vietnam's EV strategy is inseparable from VinFast, the EV arm of Vingroup conglomerate founded by billionaire Pham Nhat Vuong. VinFast is not merely a car company — it is the instrument of Vietnam's industrial ambition, the national champion vehicle through which the state pursues its aspiration to graduate from contract electronics manufacturing to high-technology automotive production.

The 2025 performance established VinFast as a serious production actor. VinFast delivered 175,099 EVs domestically in Vietnam in 2025, more than doubling from approximately 87,000 units in 2024. [Source: VinFast official release — <https://vinfastauto.us/investor-relations/news/vinfast-set-s-record-with-175099-electric-vehicles-delivered-in-vietnam-in>] The Hai Phong plant rolled out its 200,000th vehicle in 2025, with a daily production record of 1,062 units achieved in December. By H1 2026, cumulative global deliveries reached 128,662 vehicles — a 78 percent increase year-on-year — and VinFast targets 300,000 global deliveries for full-year 2026. [Source: Gulf News — <https://gulfnews.com/amp/story/business/energy/vinfasts-breakout-year-vietnamese-ev-maker-n-early-doubles-sales-in-2025-eyes-300000-deliveries-in-2026-1.500497570>]

The first seven months of 2026 saw 137,697 deliveries, already equivalent to 79 percent of VinFast's full-year 2025 domestic total — suggesting the 300,000 target is achievable.

US Listing and Capital Markets Position

VinFast listed on the Nasdaq Global Select Market under ticker VFS on August 15, 2023, through a business combination with Black Spade Acquisition Co. [Source: VinFast newsroom — <https://vinfastauto.us/newsroom/press-release/vinfast-debuts-on-nasdaq-global-select-market-following-successful-business>] The Nasdaq listing provides access to US capital markets for growth funding and establishes a public market valuation benchmark. In late June 2026, VinFast completed the divestment of its equity interest in VinFast Trading and Production JSC as part of a strategic restructuring toward a more capital-efficient, asset-light operating model — signalling that the company is prioritising manufacturing scale over vertical diversification.

BYD's Vietnam Commitment

Vietnam has attracted BYD investment through two channels. First, BYD's electronics manufacturing presence: the company's Phu Tho province facility expanded investment to \$480 million as of April 2026. [Source: TNGlobal, April 2026 — <https://technode.global/2026/04/20/chinas-byd-increases-investment-in-vietnams-northern-province-to-480m/>] Second, a new battery factory: BYD and Vietnamese automaker Kim Long Motor broke ground on a joint \$130 million battery plant on January 27, 2026, targeting 6 GWh combined annual capacity across two phases, primarily supplying commercial vehicle batteries. [Source: CnEVPost, January 2026 — <https://cnevpost.com/2026/01/27/byd-begins-construction-of-vietnam-battery-plant/>] BYD surpassed 5,000 vehicles sold in Vietnam by February 2026. [Source: CnEVPost — <https://cnevpost.com/2026/02/10/byd-surpasses-5000-sales-vietnam/>]

Vietnam's Incentive Framework

Vietnam's EV incentive policy centres on multi-year registration-fee waivers that substantially reduce the effective purchase price for consumers. Decree 182 is the governing regulatory instrument, creating a framework for FDI incentive stacking. [Source: GlobalDeal ASEAN FDI Roundup 2026 — <https://globaldeal.io/finder/articles/asean-fdi-incentives-roundup-2026>] Vietnam and Thailand are consistently cited as attracting the most EV-related FDI attention in Southeast Asia. [Source: Vietnam.vn —

<https://www.vietnam.vn/en/viet-nam-va-thai-lan-gay-chu-y-nhat-dong-nam-a-ve-xe-dien>

However, Vietnam's formal EV policy architecture is less sophisticated than Thailand's BOI system, and the government has been slower to articulate a comprehensive supply chain development strategy — relying instead on VinFast's vertical ambitions and opportunistic FDI attraction.

Charging Infrastructure: Vietnam's Structural Gap

V-Green, the charging infrastructure company launched by VinFast founder Vuong, has built an extraordinarily dense charging network. By March 2025, V-Green had constructed more than 150,000 charging points across Vietnam. [Source: Electrive — <https://www.electrive.com/2026/03/20/v-green-to-build-99-fast-charging-stations-in-vietnam/>] In March 2026, V-Green announced a 10 trillion Vietnamese Dong (\$380 million) investment to develop 99 additional fast-charging highway hubs by end-2026, each with up to 100 charging points at 150 kW capacity. [Source: Electrive, March 2026 — <https://www.electrive.com/2026/03/20/v-green-to-build-99-fast-charging-stations-in-vietnam/>] The company targets 500,000 charging points by 2028.

Vietnam's charging infrastructure per EV on road is therefore more comprehensive than Indonesia's and arguably more systematically planned than Thailand's — but it is vertically integrated into the VinFast ecosystem, which raises questions about open access for competing EV brands. Vietnam had established 28 EV charging standards and regulations by end-2025, with the full legal framework expected in Q3 2026.

The Export Challenge

Vietnam's EV ambitions are predominantly domestic-oriented. VinFast's US market expansion — dealerships and sales in California, Canada, and Europe — has been slower than originally projected, and as of mid-2026 the export-revenue contribution remains modest relative to domestic sales. Vietnam's geographic position provides reasonable logistics access to Southeast Asian neighbours but less inherent advantage for Middle East or Australian markets compared to Thailand's established export infrastructure.

VI. Head-to-Head Comparison

Competitive Matrix: Thailand vs Indonesia vs Vietnam

Factor	Thailand	Indonesia	Vietnam
Existing automotive base	Strongest in ASEAN — 1,800 parts suppliers, 1.7M+ unit annual output, 50-year OEM relationships, established export corridors	Moderate — large domestic assembly base but limited precision manufacturing depth	Weakest — predominantly contract electronics manufacturing; VinFast is the sole serious automotive actor

Factor	Thailand	Indonesia	Vietnam
Battery supply chain	Assembly-only; no significant cell production; dependent on imported Chinese packs; BOI push for cathode/cell FDI ongoing	Strongest long-term potential — world's largest nickel reserve; CATL \$6B integration project (6.9 GWh by end-2026); Hyundai-LG 10 GWh cell plant operational	Nascent; BYD-Kim Long \$130M battery plant started Jan 2026; V-Green BESS development; no cell manufacturing at scale
EV incentive regime	Most sophisticated in region — EV3.5 with 2-to-1/3-to-1 localisation ratios; hybrid extension to retain Japanese OEMs; export credit flexibility since mid-2025; \$4.1B FDI secured	VAT cut to 1% for qualifying local EVs; luxury tax exemption; import duty waivers linked to local production commitments; nickel royalty structure supports upstream integration	Decree 182 framework; registration-fee waivers; VinFast-aligned ecosystem; less institutionalised than Thailand BOI; EV standards framework completing Q3 2026
Market size	3rd largest ASEAN auto market by volume; approximately 800K-900K units/year total; EV penetration 22-30% H1 2026	Largest ASEAN auto market — 270M population; domestic demand absorbs largest EV volume if penetration scales; EV penetration still developing but Q2 2026 surged 34%	Smaller domestic market but fastest-growing EV penetration rate; EV penetration 35% H1 2026 driven by VinFast dominance; 100M+ population
Export access	Best export infrastructure in ASEAN — exports >50% of all vehicle production; established logistics to Australia, Middle East; ASEAN FTA network	Largest domestic market but export infrastructure less developed; BYD Indonesia plant targets ASEAN/ME/Africa RHD exports; FEOC complication limits US market access	Weakest export infrastructure for automotive; VinFast US/EU sales nascent; geography less advantageous for target markets
Chinese FDI	Deepest — BYD, GWM, SAIC, Changan, EV Primus all operating; 600,000+ unit capacity committed; \$4.1B total EV chain	Growing fast — BYD \$1.3B plant operational Jan 2026; CATL \$6B integration project groundbreaking June 2026; Chinese OEMs strategically important for nickel value chain	Increasing — BYD \$480M electronics and \$130M battery; Kim Long Motor partnership; BYD surpassing 5,000 sales by Feb 2026
Infrastructure	Most developed — 3,720 stations, 11,622 connectors including 6,000+ fast chargers by March 2025; targeting 12,000-station network; stricter standards from April 2026	Least developed — fewer than 5,000 public charging stations as of Dec 2025; targeting 30,000 by 2030; electricity reliability challenges remain	Dense but captive — V-Green 150,000+ charging points by March 2025; 99 new highway fast-charging hubs by end-2026; VinFast-ecosystem integration raises interoperability questions
Verdict	Current leader on breadth — deepest manufacturing ecosystem, best export infrastructure, most sophisticated policy, but risks becoming a Chinese OEM assembly platform without cell production	Best long-term structural bet — nickel-to-battery integration is unique ASEAN competitive advantage; largest domestic market; but execution risk, FEOC complications, and infrastructure gaps are real	Wild card — VinFast is a genuine national champion; charging density is impressive; but export limitations and lack of a diversified OEM base constrain the ceiling

Summary Assessment

Thailand leads on execution and breadth. Indonesia leads on structural potential. Vietnam leads on domestic EV penetration speed. The divergence reflects each nation's comparative advantages and limitations. Thailand has monetised its incumbent position effectively but risks remaining an assembly-for-export platform for Chinese OEMs rather than capturing battery-cell value. Indonesia has the most theoretically defensible supply chain position but faces execution risks and US market access complications. Vietnam has the most concentrated national champion bet — VinFast — but that concentration is both its strength and its strategic vulnerability.

VII. The Chinese OEM Strategy: ASEAN as Global Springboard

Why All Three Simultaneously

The counterintuitive feature of Chinese OEM behaviour in ASEAN is that they are not choosing between Thailand, Indonesia, and Vietnam — they are building in all three simultaneously. This is not inefficiency; it is strategic optionality. BYD has active operations in all three countries. CATL has its \$6 billion Indonesia integration project while sourcing and selling across ASEAN. GWM, SAIC, and Changan are concentrated in Thailand but exploring ASEAN-wide supply chain partnerships.

The logic is threefold. First, regulatory insurance: no single ASEAN government's policy can derail the China-to-ASEAN production migration if multiple production bases exist. Second, market access arbitrage: different ASEAN countries provide access to different downstream markets under AFTA and bilateral FTA frameworks. Thailand's EV exports can reach Australia efficiently; Indonesia's can target ASEAN neighbours under RCEP; Vietnam's location gives access to ASEAN and potentially South Asia. Third, localisation optionality: as ASEAN governments tighten local content requirements, having manufacturing presence in multiple jurisdictions allows OEMs to optimise their production splits across borders.

ASEAN as Tariff-Bypass Platform

The strategic context that makes ASEAN uniquely valuable to Chinese OEMs in 2026 is the tariff wall they face in the US, EU, Canada, Turkey, and Brazil. [Source: Rhodium Group / International News — <https://www.internationalnewsandviews.com/chinese-ev-makers-consolidation-2026-overcapacity-subsidies-end-393798-2/>] An ASEAN-manufactured vehicle with sufficient local content qualifies under AFTA rules of origin as an ASEAN product — bypassing Chinese-origin tariffs. BYD's Indonesia plant targeting 50-60 percent localisation, and its Thailand facility targeting export to ASEAN and the Middle East, are explicit implementations of this logic. Australia and Saudi Arabia, the two largest ASEAN automotive export destinations outside the region, do not currently apply Chinese-specific EV tariffs, making them natural endpoints for ASEAN-manufactured Chinese brand vehicles.

The Price-to-Quality Advantage

Chinese OEMs' competitive edge in ASEAN is not solely about price. In H1 2026 data, BYD captured 5.3 percent of Indonesia's overall vehicle market — not just the EV segment. [Source: KR-Asia — <https://kr-asia.com/ev-boom-drives-asean-car-sales-in-q2-with-indonesia-surgin-34/>] In Thailand's EV segment, 85 percent of sales are Chinese-manufactured. The quality of Chinese EVs — particularly BYD's blade battery technology, connected car features, and over-the-air software update capability — is competitive with Korean and Japanese brands at a 20-30 percent price discount in comparable segments. Vietnamese consumers face a similar value proposition from BYD vs VinFast, and VinFast's domestic market dominance (35 percent EV penetration) is partially

explained by VinFast's aggressive pricing strategy and V-Green's captive charging ecosystem.

The Overcapacity Glut Risk

The strategic risk in the Chinese ASEAN playbook is that production capacity in ASEAN could significantly exceed regional demand absorption. Of approximately 130 Chinese EV companies, only three — BYD, Li Auto, and Geely — were recording profits as of mid-2025. [Source: International News and Views, 2026 — <https://www.internationalnewsandviews.com/chinese-ev-makers-consolidation-2026-overcapacity-subsidies-end-393798-2/>] China's total vehicle production capacity stands at around 50 million units against projected output of 33 million in 2025 — a massive structural overcapacity. As Chinese manufacturers redirect surplus production to ASEAN, the risk of an EV glut — depressed prices, market disruption, and host-country trade defence responses — is real. Indonesia, Malaysia, Thailand, and Vietnam are already applying or considering antidumping duties on Chinese steel exports that have surged into the region. A similar trade protection response to EV dumping is not inconceivable by 2028-2030.

VIII. Japan's Last Stand: Hybrid Dominance in a Transitioning Market

Toyota's Multi-Pathway Defence

Japan's automotive giants — Toyota, Honda, Mitsubishi, Suzuki — have dominated ASEAN vehicle markets for decades. Thailand alone produced millions of Japanese-branded vehicles annually before the Chinese EV wave. The strategic response from Japanese OEMs to Chinese EV displacement is crystallised in Toyota's "multi-pathway strategy": the argument that diverse powertrains — hybrid, plug-in hybrid, hydrogen, and full BEV — each have a role in market-specific decarbonisation pathways, and that betting exclusively on BEV is premature.

Toyota cut its 2026 BEV production target by 30 percent to one million units globally while projecting hybrid production surges to 5 million units annually. [Source: Gulf News / Honda Bloomberg — <https://gulfnews.com/amp/gulfnews/auto/news/toyota-to-sell-15-million-evs-by-2026-roll-out-10-new-models-1.94991296>] In fiscal 2025, Toyota's global BEV sales increased 30 percent to 188,000 units — against a total electrified vehicle fleet including hybrids of approximately 4.7 million. [Source: Japan Times — <https://www.japantimes.co.jp/environment/2026/05/24/sustainability/auto-industry-decarbonization/>] Honda is targeting 1.3 million hybrid sales annually by 2030, doubling down on hybrid technology with two new production platforms launching in 2026.

The ASEAN Hybrid Stronghold

In ASEAN's current market reality, Japanese hybrid dominance has a strong rationale. The charging infrastructure gaps in Indonesia, Malaysia, the Philippines, and parts of Vietnam make BEV range anxiety a genuine consumer barrier. Hybrids — particularly Toyota's self-charging hybrid system — require no charging infrastructure and deliver fuel economy improvements of 30-40 percent over

conventional combustion vehicles. In price-sensitive ASEAN markets, this value proposition remains compelling.

Thailand's December 2025 extension of BOI incentives to hybrid and mild-hybrid vehicles is a direct outcome of Japanese lobbying and represents a genuine policy achievement for Toyota, Honda, and Mitsubishi. The EV3.5 expansion to hybrids provides a production incentive structure that Japanese OEMs can utilise without committing to full BEV manufacturing in Thailand. The Megatech Thailand analysis describes Japanese automakers as "executing a pragmatic pivot back to hybrids and plug-in hybrids, capitalising on their historical dominance." [Source: Megatech Thailand — <https://megatechthailand.com/jp/manufacturing-trends/japanese-automakers-in-the-global-race-toward-electrification/>]

When Does the Hybrid Defence Fail?

The durability of Japan's ASEAN hybrid strategy depends on three variables: the pace of charging infrastructure build-out, the price convergence between Chinese BEVs and Japanese hybrids, and ASEAN government policy evolution. On all three vectors, the trend is adverse for Japanese OEMs. Thailand is building toward 12,000 charging stations; Vietnam has 150,000 charging points; Indonesia targets 30,000 public stations by 2030.

Chinese BEV prices are declining structurally due to battery cost reductions — LFP battery pack costs have fallen below \$100/kWh — while Japanese hybrid prices remain premium. And ASEAN governments are consistently tightening ZEV targets: Thailand's 30@30 strategy explicitly disadvantages pure hybrid production beyond 2027.

The consensus view among automotive analysts is that Japanese OEMs have until approximately 2028-2029 before their hybrid buffer erodes in Thailand and Vietnam. In Indonesia, the timeline is slightly longer due to infrastructure gaps — possibly 2029-2031. By 2030, Japanese brand dominance in ASEAN will likely have shifted from majority-market to significant-minority position in all three countries studied here, unless Japanese OEMs accelerate BEV investment substantially beyond current plans.

IX. Superforecast — The Winner by 2030

The Scoring Framework

Forecasting a winner in a three-way geopolitical-industrial competition requires explicit criteria. This report assesses five dimensions over the 2026-2030 horizon: (1) absolute EV production volume; (2) battery supply chain integration depth; (3) EV export volume and market diversity; (4) domestic brand development; and (5) policy ecosystem stability and sophistication.

Thailand's Trajectory: First Mover Advantage, Structural Ceiling

Thailand's probability of being ranked the leading ASEAN EV hub by 2030 is assessed at approximately 45 percent. The case rests on execution momentum: \$4.1 billion in committed EV supply chain FDI, operational Chinese OEM assembly plants with 600,000+ unit combined capacity, the best export infrastructure in ASEAN, and the most sophisticated policy architecture. If Thai BOI succeeds in attracting battery cell manufacturers to complement its assembly base, Thailand's lead becomes durable.

The structural ceiling risk is real: without cell production, Thailand risks becoming an assembly platform for Chinese OEMs whose loyalty is to their own global supply chain optimisation rather than Thailand's national industrial interests. The hybrid policy extension to retain Japanese OEMs adds investment breadth but may inadvertently slow the pure-EV supply chain deepening that is required to win the next competitive phase.

Indonesia's Trajectory: The Long Game, High Variance

Indonesia's probability of leading by 2030 is assessed at approximately 35 percent. The nickel-to-battery vertical integration thesis is the most compelling structural argument in ASEAN — if executed, it represents a defensible moat that no other ASEAN nation can replicate. The CATL \$6B project and Hyundai-LG 10 GWh cell plant are genuine foundations, not aspirational announcements.

However, high variance surrounds the Indonesia forecast. The FEOC complication limits US market access. Infrastructure gaps — fewer than 5,000 public charging stations versus Thailand's 11,600-plus — constrain domestic demand growth. And CATL's history of protecting core battery technology from transfer creates a risk that Indonesia hosts the capital investment but not the intellectual property. Prabowo's resource nationalism via royalty increases demonstrates political confidence, but also risk of deterring the very FDI the vertical integration thesis requires.

Vietnam's Trajectory: The Concentrated Bet

Vietnam's probability of leading by 2030 is assessed at approximately 15 percent. VinFast's trajectory is impressive — 200,000 production units in 2025, targeting 300,000 deliveries in 2026, with 150,000 charging points nationwide — but Vietnam's EV leadership is structurally dependent on a single corporate actor. VinFast's US expansion difficulties, the lack of a second major automotive investor, and the absence of battery cell manufacturing constrain Vietnam's ceiling.

Vietnam's 35 percent EV penetration rate is the highest in ASEAN and represents a genuine achievement — but it reflects VinFast's domestic market monopoly more than a diversified EV industrial ecosystem. If VinFast stumbles — in US market execution, in manufacturing quality, or in financial management post-Nasdaq listing — Vietnam has no fallback manufacturer. This concentration risk is Vietnam's fundamental strategic vulnerability.

The Most Likely Outcome: A Two-Hub ASEAN

The most probable outcome by 2030 is not a single winner but a two-hub configuration. Thailand leads on assembly, exports, and policy sophistication — the "manufacturing platform" hub. Indonesia leads on battery supply chain integration — the "resource-to-battery" hub. The two complementary roles create an ASEAN supply chain structure in which Thai assembly plants draw on Indonesian battery cells and modules, with Chinese OEMs orchestrating the cross-border supply chain. Vietnam operates as a third, smaller hub — domestically significant through VinFast but not an ASEAN-scale producer.

Japan retains 30-40 percent of total ASEAN vehicle production by 2030 through hybrid product lines and long-established supply chain relationships, but its share of the specifically EV segment shrinks dramatically toward a 10-20 percent range.

The geopolitical wildcard that could reshape this forecast is US-ASEAN trade policy on FEOC compliance and rules-of-origin certification for IRA credits. If Indonesia resolves its FEOC complication through a bilateral critical minerals agreement with Washington, its strategic position improves substantially and the probability distribution shifts toward Indonesia leading the integrated battery-to-assembly arc. If US tariffs extend to ASEAN-made Chinese-brand vehicles — closing the export bypass route — Thailand's assembly-for-export model faces a demand shock and its probability weighting declines.

The race is not over. But the shape of the outcome is becoming visible.

X. Data Sources

All market figures in this report are sourced from live web searches conducted in August 2026. No training data has been used for any quantitative claim.

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